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Subject:	Distributed Computing
Experiment No.:	3
Project title:	Travel Itinerary Planner
Problem Statement:	Implementation of Multithreading in Distributed System
Objectives:	To implement a multithreaded server that handles multiple client requests simultaneously and applies multithreading to fetch different itinerary components (restaurants, events, attractions, and transport) in parallel for faster and more responsive travel planning.
Theory:	**What is Multithreading?** In a distributed system, multithreading is a software design pattern where a single process on one or more networked computers is divided into multiple, concurrent threads of execution that can perform tasks simultaneously. These threads operate within the same process and share its resources, such as memory and I/O, allowing for efficient parallel processing, improved application responsiveness, and higher throughput across the distributed system. **Benefits of Multithreading in Distributed Systems:** 1. Better performance: Tasks can run at the same time (parallel), so the system responds faster. 2. Efficient resource use: CPU and memory are used properly instead of sitting idle. 3. Faster response time: Even if one task is waiting (e.g., for data), others can continue running. 4. Handles many users: Multiple users can be served at once without slowdown. 5. Improved scalability: Easy to add more tasks/threads as the system grows. 6. Reduced waiting time: Long tasks (such as fetching large data) don't block smaller, quicker tasks. 7. Supports real-time processing: Useful for systems needing

quick updates (e.g., live info).

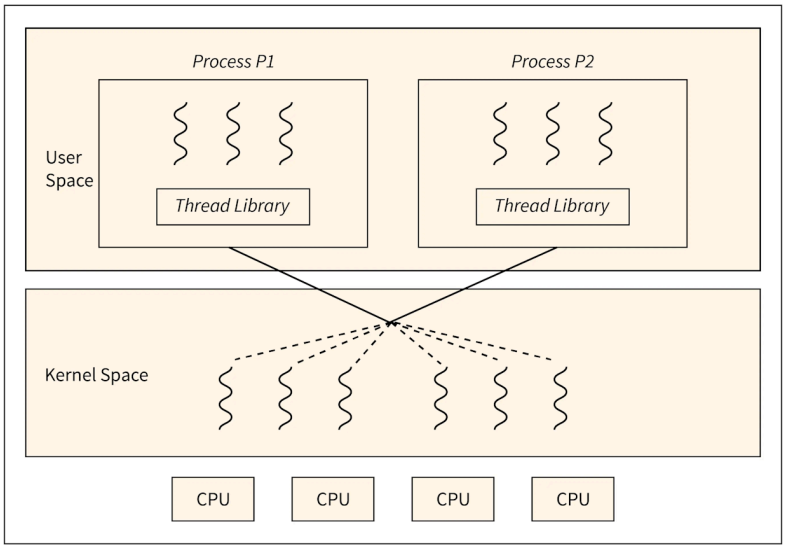


Figure 1: Multithreading architecture showing user-level threads managed by the thread library and mapped to kernel threads for execution on multiple CPUs.

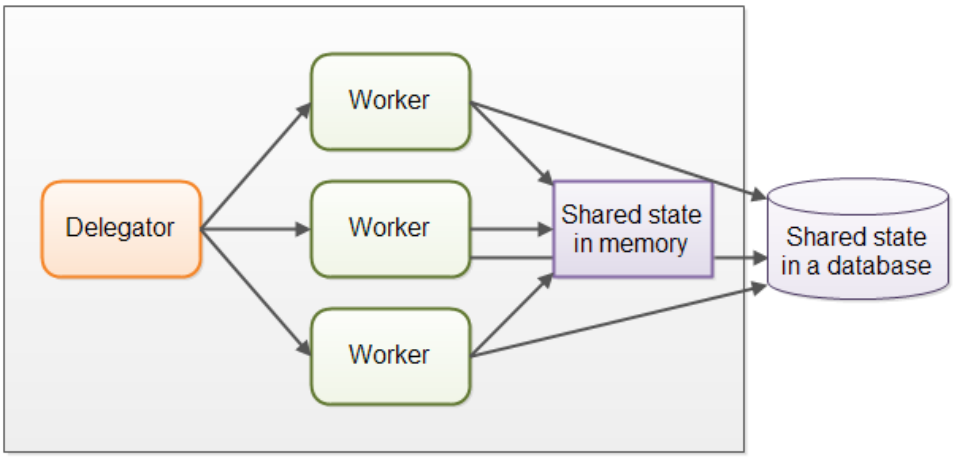


Figure 2: A Fine-Grain Multithreading Superscalar Architecture by Loikkanen and Bagherzadeh:

	In our Travel Itinerary Planner server, multithreading enables the system to handle multiple client connections and multiple services simultaneously. Each client request runs in its own thread. This ensures that many users can interact with the system at the same time without waiting for others, improving responsiveness and scalability. For example: ● One thread can fetch restaurants from the database. ● Another thread can retrieve events happening at the same location. ● A separate thread can list attractions nearby. ● Another thread can check transport options between locations. Because these threads run in parallel, itinerary generation becomes much faster compared to sequential execution. Even if fetching transport data takes longer, other services (like restaurants or attractions) are not blocked. Multithreading also improves the user experience: ● While one user logs in, another can search for events. ● A third user can plan transport options without affecting the others. ● Background threads keep updating data (such as live transport or event availability) while the main thread handles user interactions. Thus, with multithreading, my system supports real-time, distributed itinerary planning where restaurants, events, attractions, and transport information are all processed simultaneously and combined into a complete travel plan.

Code:	**Server:-**
	`import threading`
	`import socket`
	`import time`
	`import json`
	`import sqlite3`
	`from datetime import datetime`
	
	`HOST = "127.0.0.1"`
	`PORT = 5000`
	
	`def client_request(conn, addr):`
	`  print(f"[NEW CONNECTION] {addr}")`
	`  print(f"Active threads: {threading.active_count() -`

```
1}")
    data = conn.recv(1024).decode()
    msg = json.loads(data)
    if msg.get("req") == "restaurants":
        location = msg.get("location", "Goa")
        min_price = msg.get("min", 0)
        max_price = msg.get("max", 100000)
        db_conn = sqlite3.connect("Main_DB.db")
        curr = db_conn.cursor()
        results = curr.execute(
            "SELECT name, price FROM restaurants WHERE
location=? AND price BETWEEN ? AND ?",
            (location, min_price, max_price)
        ).fetchall()
        db_conn.close()
        time.sleep(80)
        if results:
            response = [[name, price] for name, price in
results]
            conn.send(json.dumps(response).encode())
        else:
            conn.send("No results".encode())

    elif msg.get("req") == "transport":
        curr_loc = msg.get("Current")
        dest = msg.get("Destination")
        db_conn = sqlite3.connect("Main_DB.db")
        cur = db_conn.cursor()
        results = cur.execute(
            "SELECT
start_location,end_location,airline,price FROM
transport_routes WHERE start_location=? AND end_location=?
ORDER BY price ASC",
            (curr_loc, dest)
        ).fetchall()
        db_conn.close()
        time.sleep(80)
        if results:
            response = [[start_location, end_location,
```

```
airline, price] for start_location, end_location, airline,
price in results]
            conn.send(json.dumps(response).encode())
        else:
            conn.send("No results".encode())

    elif msg.get("req") == "attractions":
        loc = msg.get("location")
        type_ = msg.get("type", "")
        db_conn = sqlite3.connect("Main_DB.db")
        cur = db_conn.cursor()
        if not type_:
            results = cur.execute(
                "SELECT name,location,category FROM
attractions WHERE location=?",
                (loc,)
            ).fetchall()
        else:
            results = cur.execute(
                "SELECT name,location,category FROM
attractions WHERE location=? AND type=?",
                (loc, type_)
            ).fetchall()
        db_conn.close()
        time.sleep(80)
        if results:
            response = [[name, location, category] for
name, location, category in results]
            conn.send(json.dumps(response).encode())
        else:
            conn.send("No results".encode())

    elif msg.get("req") == "events":
        start = msg.get("start")
        loc= msg.get("location")
        end = msg.get("end")
        start_date = datetime.strptime(start,
"%d-%m-%Y").strftime("%Y-%m-%d")
        end_date = datetime.strptime(end,
```

```
"%d-%m-%Y").strftime("%Y-%m-%d")
        db_conn = sqlite3.connect("Main_DB.db")
        cur = db_conn.cursor()
        results = cur.execute(
            "SELECT name,location,date,category FROM
events WHERE date >= ? AND date <=? AND location=?",
            (start_date, end_date,loc)
        ).fetchall()
        db_conn.close()
        time.sleep(80)
        if results:
            response = [[name, location, date, category]
for name, location, date, category in results]
            conn.send(json.dumps(response).encode())
        else:
            conn.send("No results".encode())

    conn.close()
    print(f"[DISCONNECTED] {addr}")

def start_server():
    server = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
    server.bind((HOST, PORT))
    server.listen()
    print(f"[LISTENING] on {HOST}:{PORT}")
    while True:
        conn, addr = server.accept()
        thread = threading.Thread(target=client_request,
args=(conn, addr))
        thread.start()

start_server()
```

Client:-

```
import socket
import json
```

```
def rest(loc,min_price,max_price):

client=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
    client.connect(("localhost",5000))
    data = {
        "req":"restaurants",
        "location":loc,
        "min":min_price,
        "max":max_price
    }
    client.send(json.dumps(data).encode())
    data = client.recv(1024).decode()
    print(data)
    client.close()

def transport(curr_loc,dest):

client=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
    client.connect(("localhost",5000))
    data = {
        "req":"transport",
        "Current":curr_loc,
        "Destination":dest
    }
    client.send(json.dumps(data).encode())
    data = client.recv(1024).decode()
    print(data)
    client.close()

def attractions(location,type:str=None):

client=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
    client.connect(("localhost",5000))
    data = {
        "req":"attractions",
        "location":location,
```

```
        "type":type if type else ""
    }
    client.send(json.dumps(data).encode())
    data = client.recv(1024).decode()
    print(data)
    client.close()

def events(loc, start,end):

client=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
    client.connect(("localhost",5000))
    data = {
        "req":"events",
        "location":loc,
        "start":start,
        "end":end
    }
    client.send(json.dumps(data).encode())
    data = client.recv(1024).decode()
    print(data)
    client.close()
ch=1
while(ch!=3):

    print('''

       Menu
       1. Restaurants/Hotels
       2. Events
       3. Transport Routes
       4. Attractions

    '''
    )

    ch=int(input("Enter your choice: "))

    if(ch==1):
        loc=str(input("Enter desired location: "))
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        max = int(input("Enter Max Price:"))
        rest(loc,0,max)

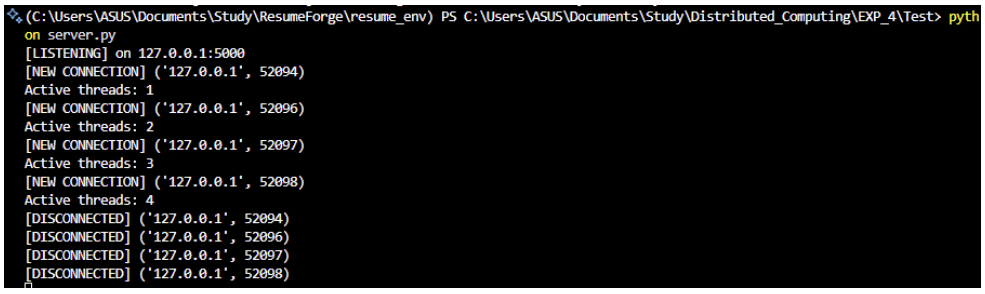

    elif(ch==2):

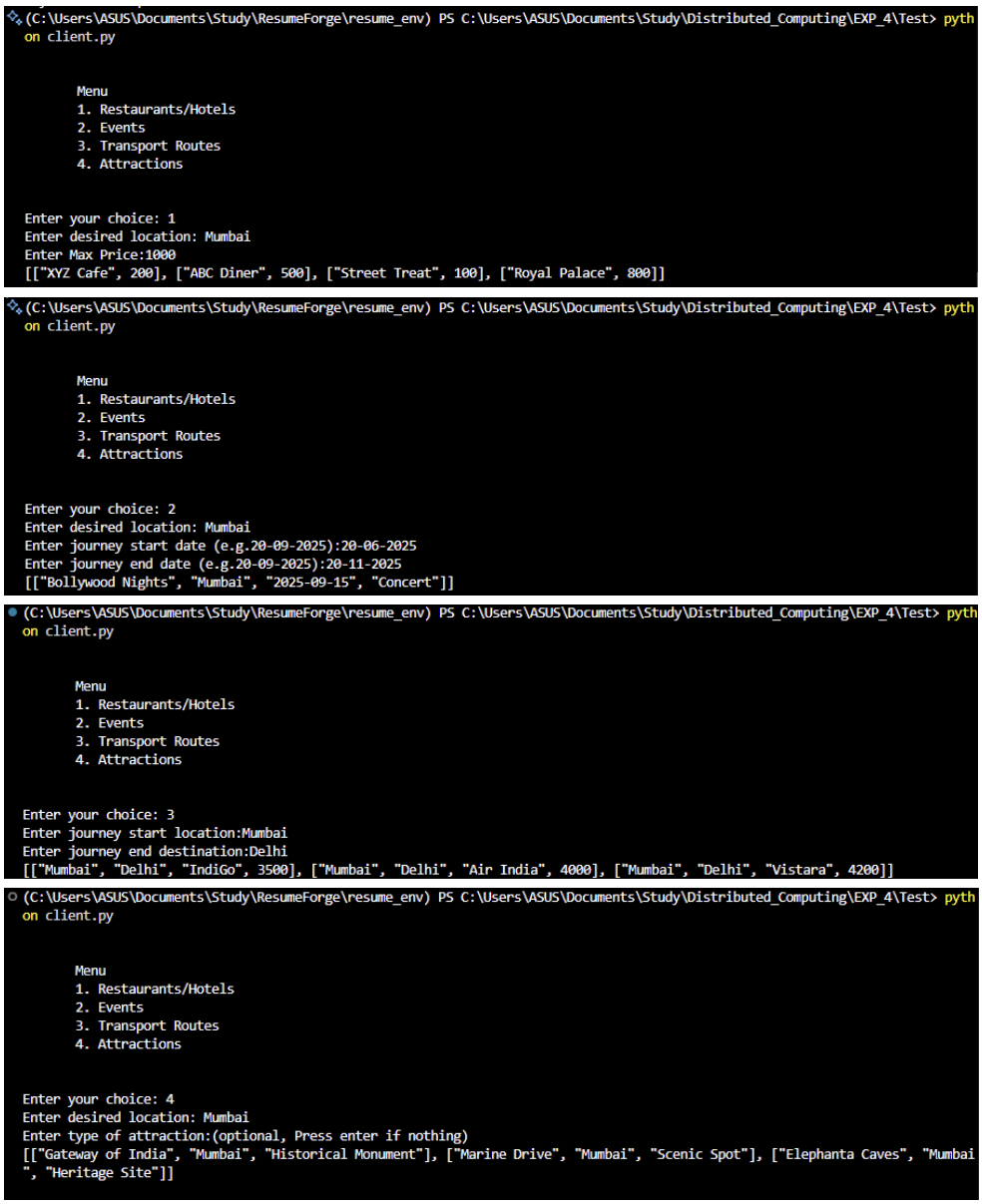
        loc=str(input("Enter desired location: "))
        start = str(input("Enter journey start date
(e.g.20-09-2025):"))
        end =str(input("Enter journey end date
(e.g.20-09-2025):"))
        events(loc,start,end)

    elif(ch==3):
        curr = str(input("Enter journey start location:"))
        dest = str(input("Enter journey end
destination:"))
        transport(curr,dest)

    elif(ch==4):
        loc=str(input("Enter desired location: "))
        type_att = str(input("Enter type of
attraction:(optional, Press enter if nothing)"))
        attractions(loc,type_att)
    else:
        print("Thank You! Bye")
```

Output:





Conclusion:	This experiment showed how multithreading makes a distributed system faster and more responsive by handling many client requests at the same time. In the travel itinerary planner, it helped fetch restaurants, events, attractions, and transport details in parallel, giving users a smooth and efficient experience.